

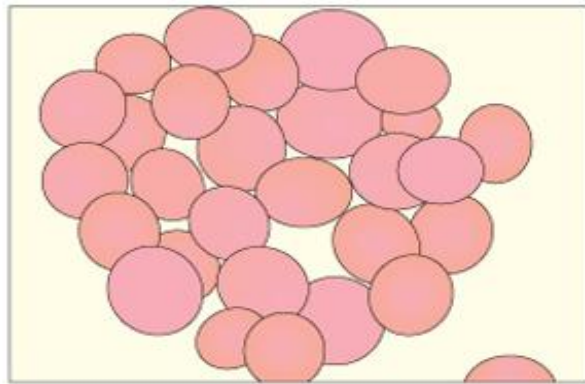
VECTOR BORNE BACTERIAL DISEASES

1. RELAPSING FEVER & LYME DISEASE

***Borrelia* spp.**

General microbiologic characteristics of genus Borrelia

- Gram-negative flexible, spiral-shaped, motile with endo/internal flagella (7-20), 0.2 to 0.5 μm wide by 4 to 25 μm long, microaerophilic
- Single Linear chromosome + Linear and circular plasmids
- The generation time is slow, about 10-12 hours
- Can be easily stained with Giemsa and Wright stain
- All borrelia are transmitted by blood-feeding arthropods (ticks/louse)



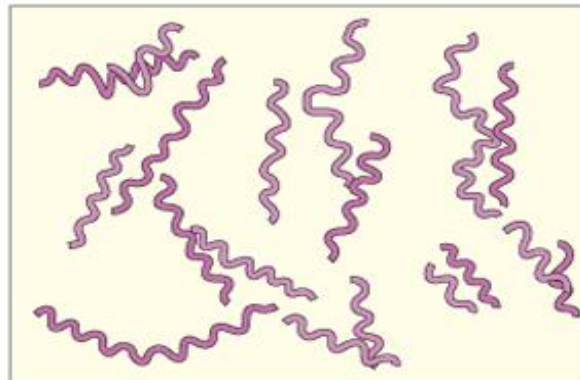
coccus/cocci



bacillus/bacilli

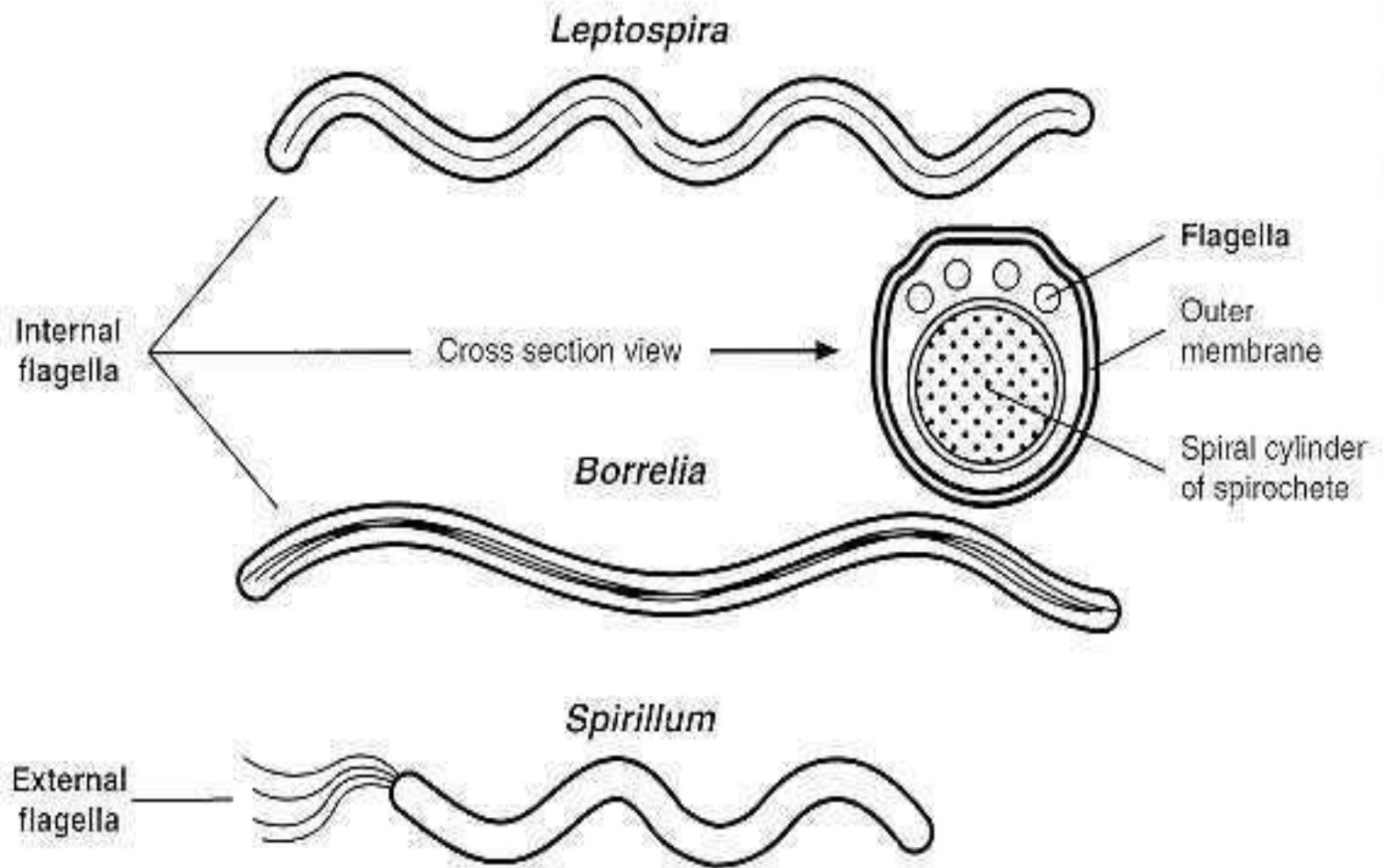
→ fusiform

→ coccobacilli



spirochetes

→ vibrios spirilla



Epidemic relapsing fever (Louse borne)

Borrelia recurrentis

Vector: *Pediculus humanus var. humanus* (human body louse)

MT: Scratching or crushing of infected louse at the site of bite/superficial wound

Person to person transmission is common through infected human body louse

Reservoir: Human

IP: 3-10 days

Risk factor: Overcrowding, poverty, poor personal hygiene

Endemic relapsing fever (Tick borne)

- *Borrelia duttonii*

Vector: *Ornithodoros* spp. (soft ticks)

MT: Tick bite

Reservoir: Human

Others

- *Borrelia hermsi*
- *Borrelia turicatae*
- *Borrelia persica*
- *Borrelia hispanica*
- *Borrelia crocidurae*

**Soft shelled ticks
(*Ornithodoros* spp.)**

Tick Bite

Reservoirs: Rodents & Ticks

Lyme disease

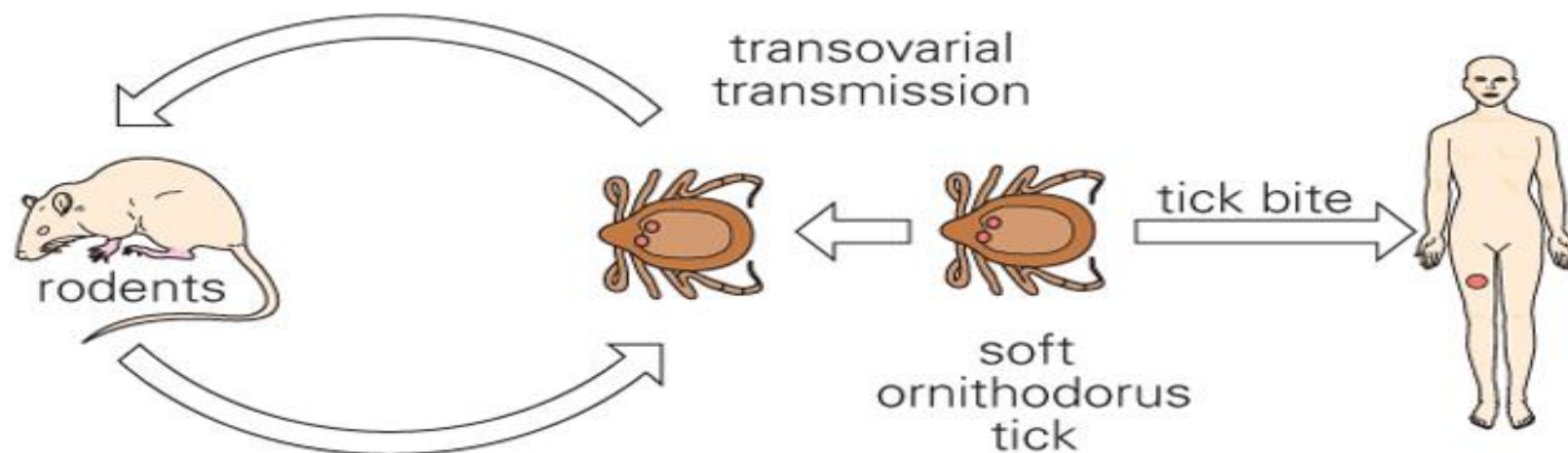
- *Borrelia burgdorferi*
- *Borrelia afzelli*
- *Borrelia garinii*

Ixodes spp. (hard shelled ticks)

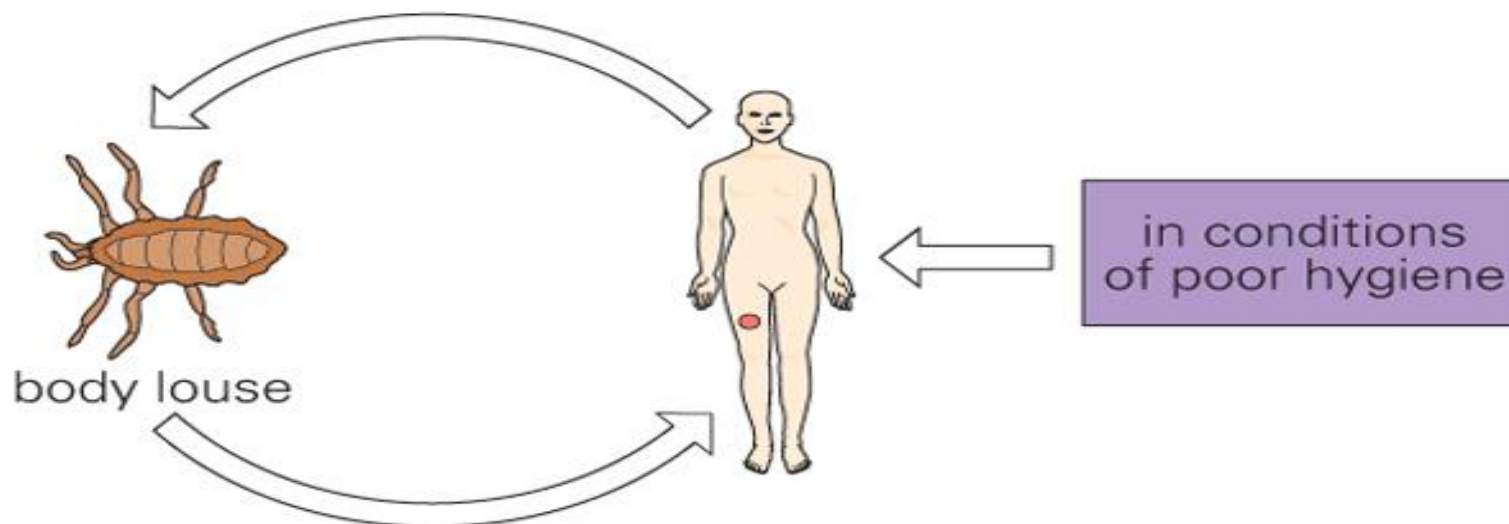
Tick Bite

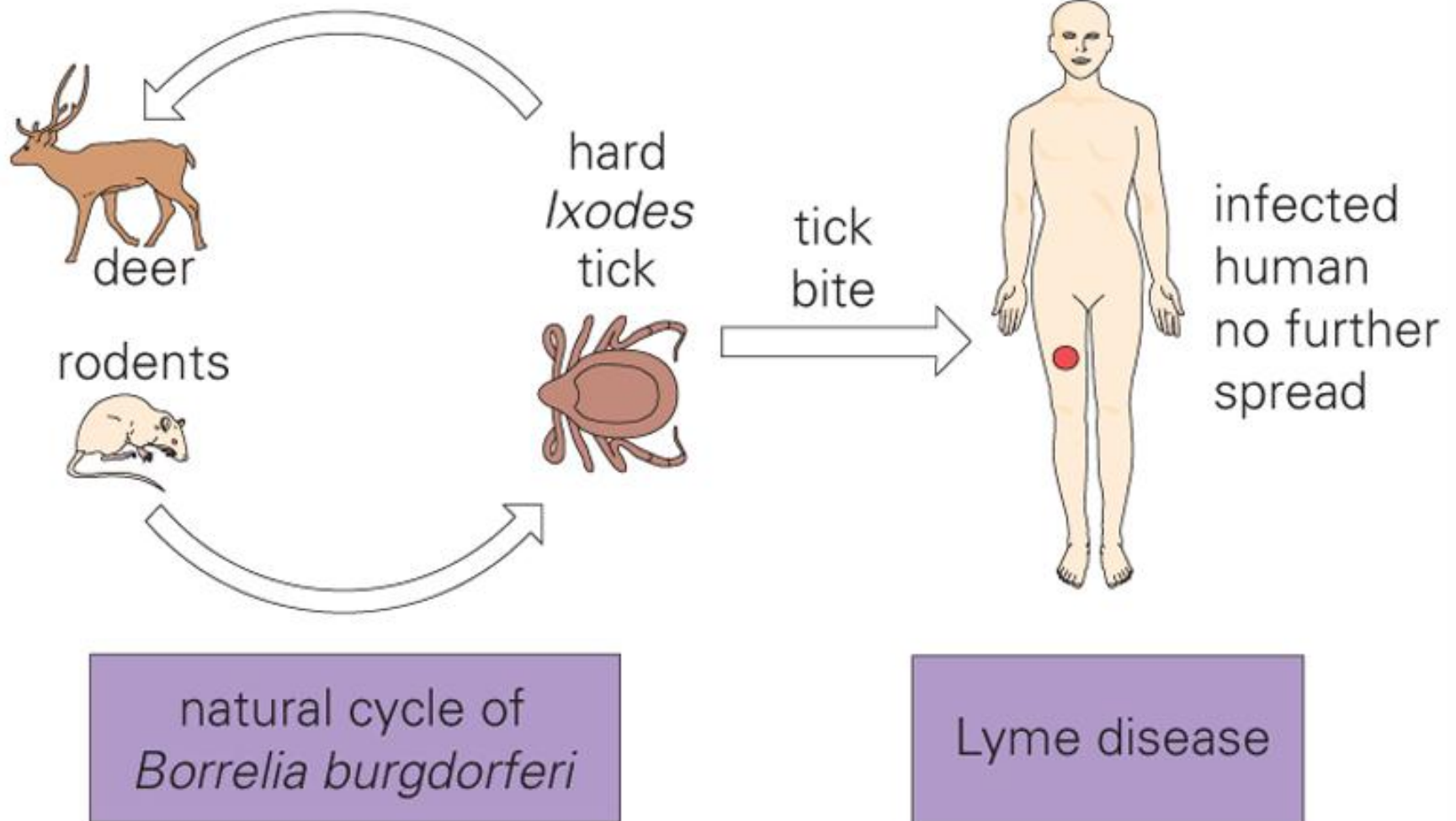
Reservoirs: Rodents, Deer, Ticks

endemic tick-borne



epidemic louse-borne





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Transmission of Lyme disease

Epidemiology of *Borrelia* Infections

Borrelia recurrentis

***Borrelia* spp.**

Borrelia burgdorferi

Infection	Reservoir	Vector
Relapsing fever Epidemic (louse-borne)	Humans	Body louse <i>Pediculus humanus</i> 
Relapsing fever Endemic (tick-borne)	Rodents, soft-shelled ticks	Soft-shelled tick <i>Ornithodoros</i> spp. 
Lyme disease	Rodents, deer, domestic pets, hard-shelled ticks	Hard-shelled tick <i>Ixodes</i> spp. 

Virulence factors

Borrelia spp.

- Outer surface proteins (Osp)-OMP
- *Borrelia recurrentis* shows antigenic variation (highly variable):
10⁴ generation
Helps the bacteria to escape from host immune response
during infection

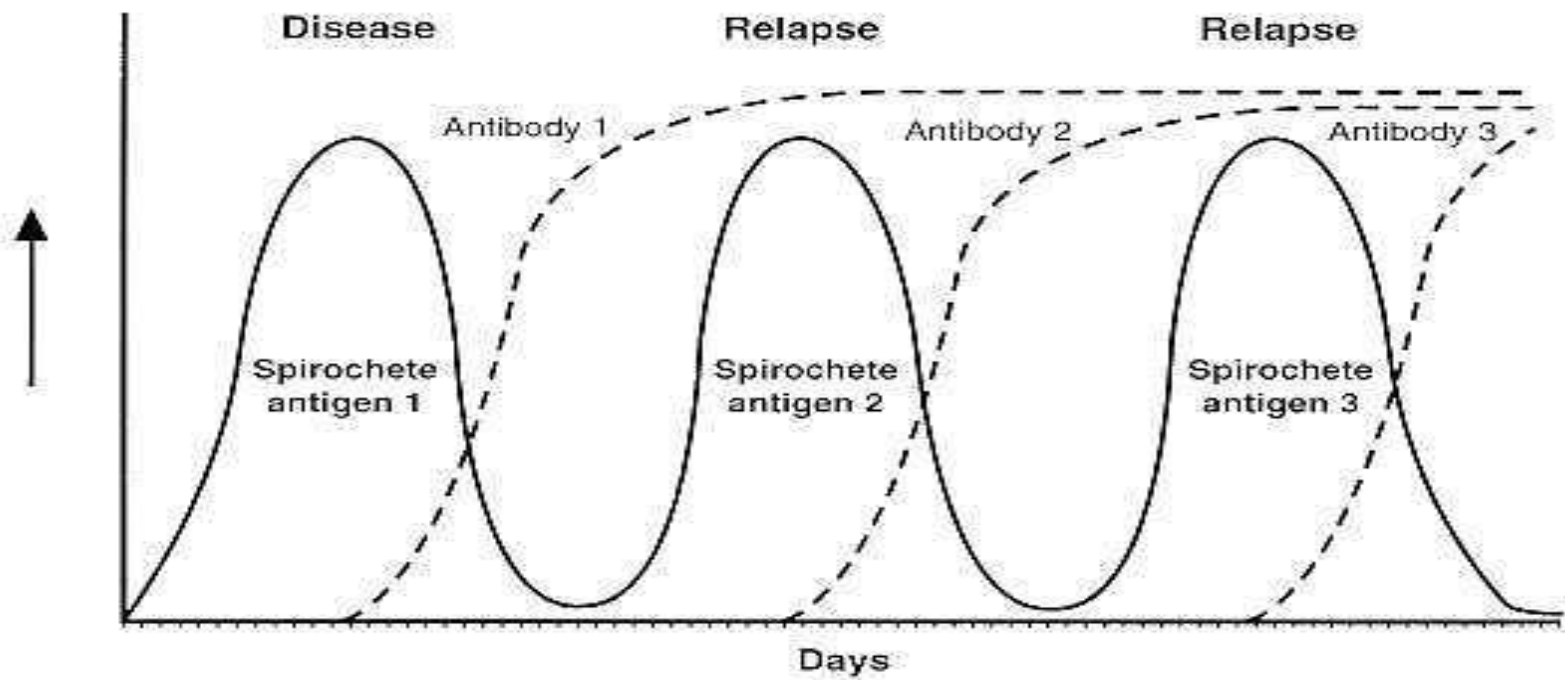
Clinical features

Louse-borne Relapsing fever

- Fever, headache, and myalgia that persist for 4-10 days, followed by an afebrile period of 5-6 days correlated with the absence of spirochetemia. A single/two relapse/s occurs in louse-borne RF
- Relapses are due to the ability of borreliae to undergo multiple cyclic antigenic variations-evade immune responses
- MR: in untreated epidemic RF can be higher than 40%, and myocarditis probably is the most common cause of death.

Tick-borne Relapsing fever

- Similar to the louse-borne RF
Less severe (mortality, 0-8 %), and several relapses (up to 13) of decreasing intensity are commonly experienced



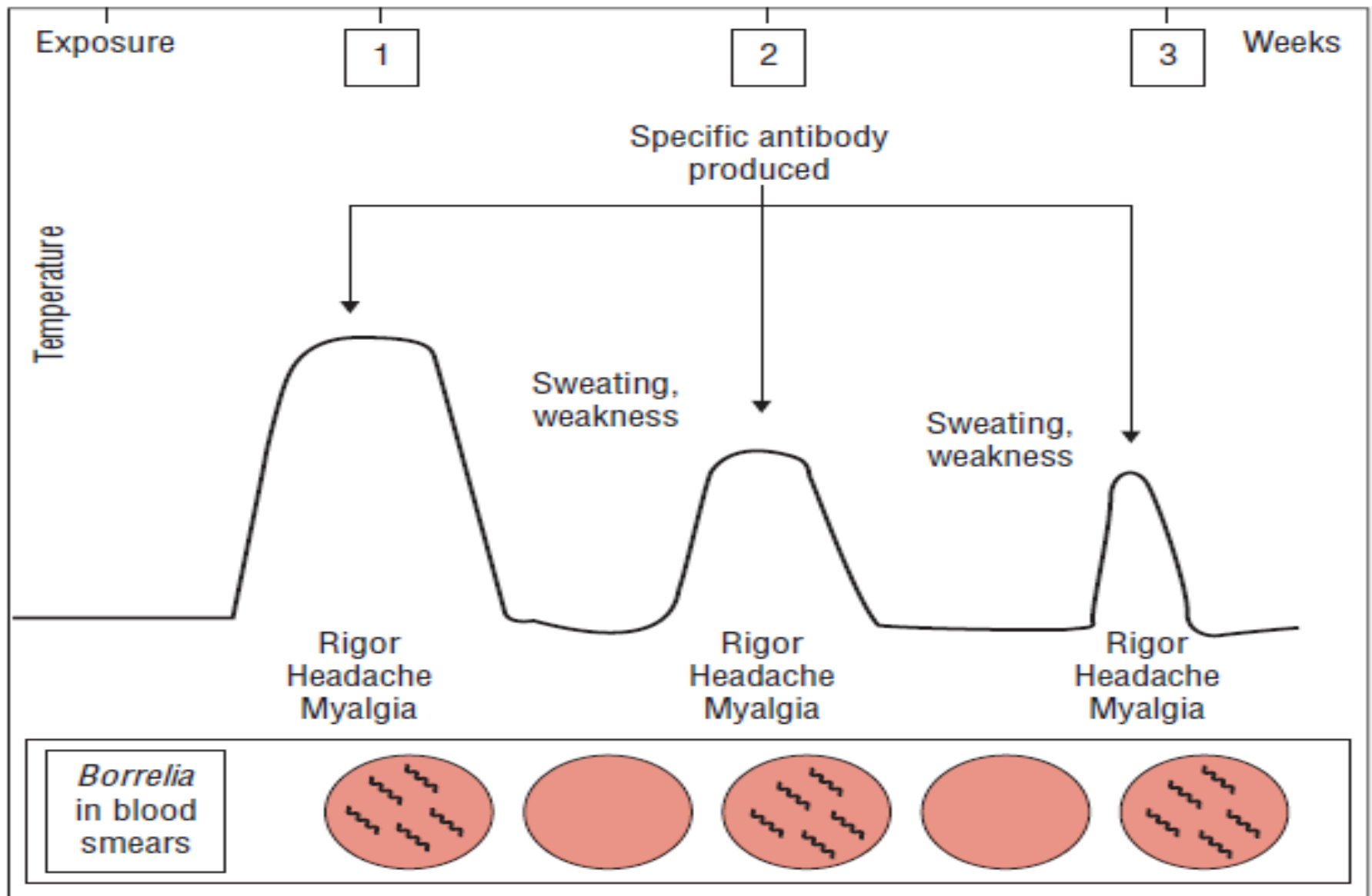


FIG. 44-4. Course of relapsing fever.

Lyme disease (Lyme borreliosis) (*Borrelia burgdorferi*)

Stage one (early infection)

- Expanding red rash that usually develops at the site of the tick bite (3 days-1month). This rash, known as erythema chronicum migrans with characteristic, "bull's eye" appearance.
- Erythema migrans persists longer, but usually subsides within 3-4 weeks.

Stage two (dissemination stage)

- Spirochetes spread hematogenously to additional body tissues
fatigues, chills, fever, headache, muscle and joint pain, swollen lymph nodes, and secondary annular skin lesions



Erythema chronicum migrans with bull eye appearance in Stage 1

Stage three (persistent infection) (Weeks, months, or years)

- o Joints, tendons, muscle and bone are affected
- o Erosion of cartilage and/or bone (deformity of joints)
- o Nervous system abnormalities can include numbness, pain, Bell's palsy and meningitis
- o Cardiac abnormalities, irregularities of the heart rhythm occur

Laboratory Diagnosis of Borrelia Infections

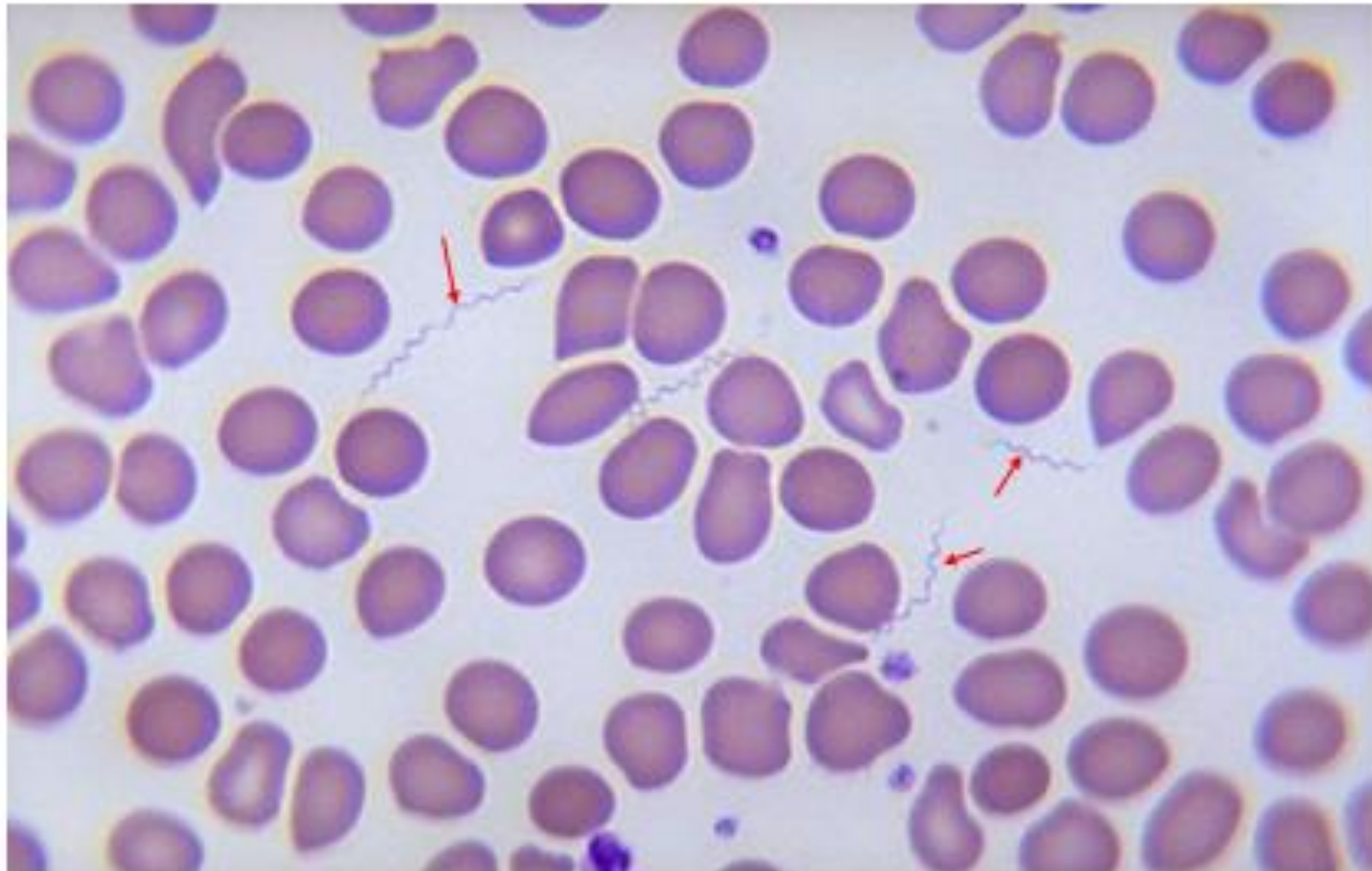
Relapsing fever

- Blood smear from febrile patient
 - Wright/Giemsa/Leishman stain
 - Microscopic examination: Observe spirochetes in the peripheral blood smear



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Giemsa stain: Tightly coiled helical spirochetes of *Borrelia recurrentis* in the blood of a patient with relapsing fever



Borrelia recurrentis-from peripheral blood film Relapsing fever (Wright stain)

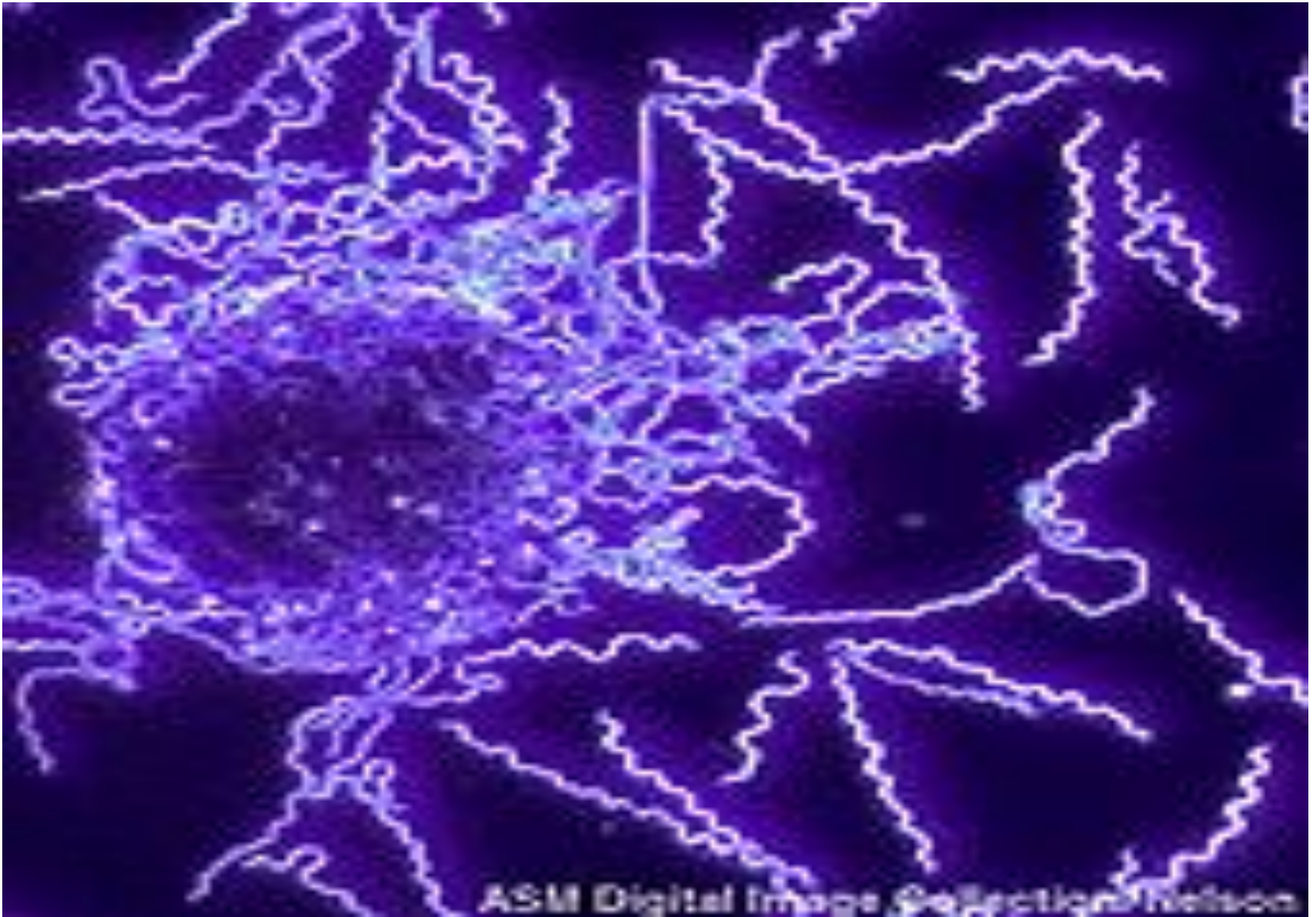
Lyme disease

Dark field microscopy

DFA

Culture

Serology (Stages 2 and 3)



Treatment

Relapsing fever

- Drugs of choice: Tetracycline/Penicillin/Erythromycin/CAF
 - JHR (Jarisch-Herxheimer Reaction): 1-2 hrs after Rx
 - MR: 1-5%



- Endotoxin released from dead spirochetes stimulates the release of cytokines (IL-6, TNF alpha, IL-8)



- Hypotension, shock, death

Prevention

- Personal hygiene: Louse borne RF
- Ticks/rodents control: Tick borne RF
- No vaccines have been developed for RF

2. TYPHUS AND OTHER RICKETTSIA DISEASES

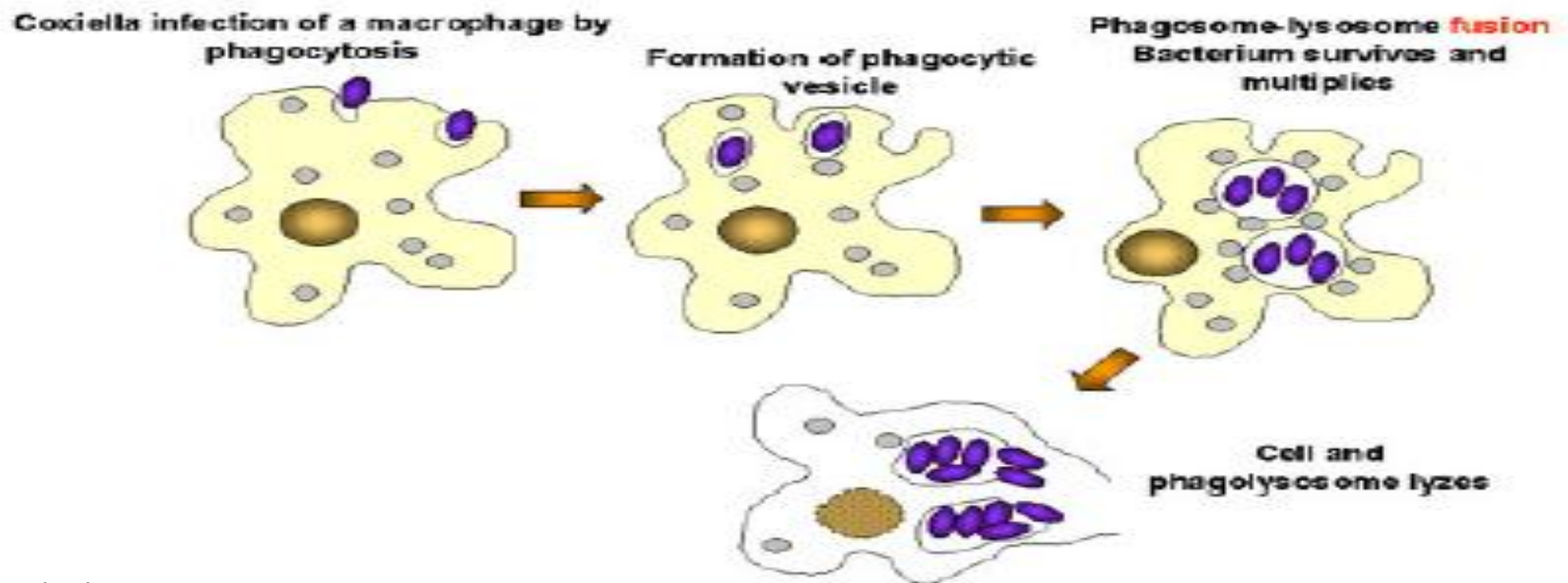
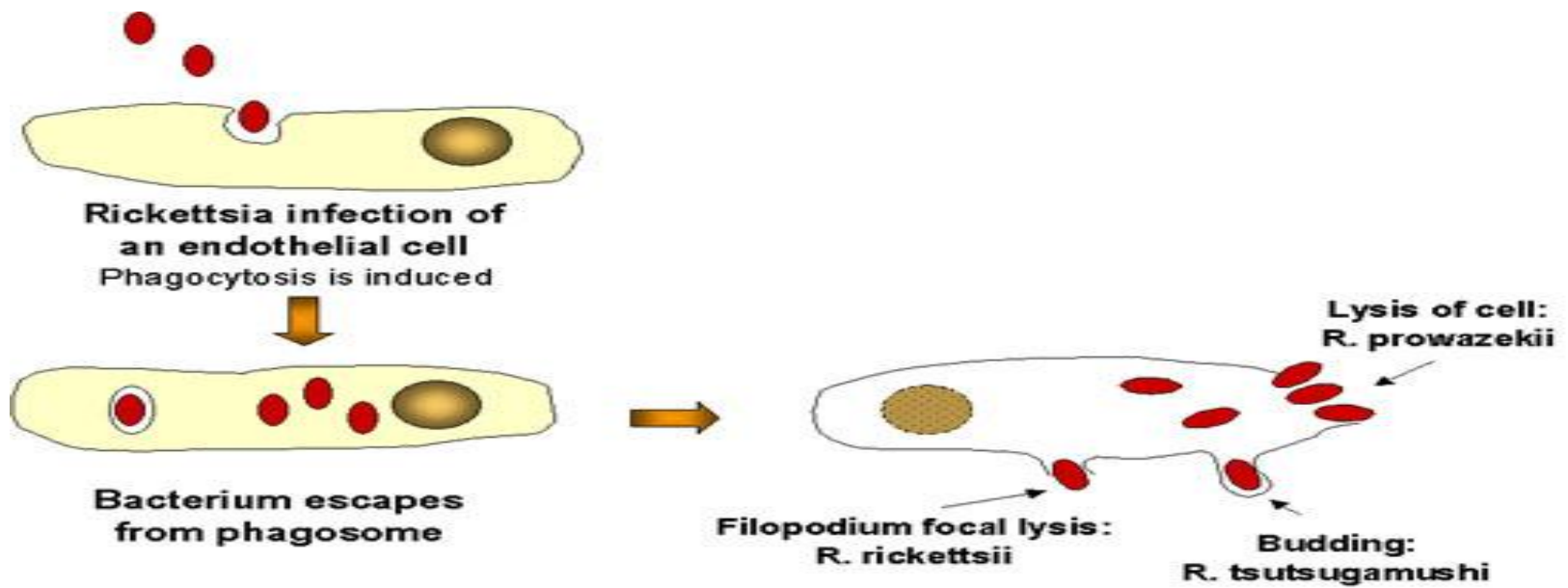
*Rickettsia, Coxiella, Orientia and
Bartonella spp.*

General microbiologic characteristics

- Small, gram negative pleomorphic cocco-bacilli obligate intracellular bacteria
- All are transmitted by arthropods (tick, mite, flea or louse) to humans except *Coxiella*
- Genus *Coxiella* has an endospore like structure and relatively resistant to desiccation, heat and sunlight, and is therefore stable enough to be acquired from infected material by inhalation

Virulence factors and pathogenesis

- LPS: Endotoxin
- Phospholipase A : Facilitate entry into the cells
- Target site: Endothelial cells of small blood vessels of any organs (skin, brain, lungs, heart, kidneys, liver, GI)
- Inflammation-----vasculitis----rash $\pm$ necrotic lesion



The Principal Rickettsia Diseases in Humans

Organism	Disease	Arthropod vector	Vertebrate reservoir	Clinical severity	Geographic distribution
I. Typhus					
<i>R. prowazekii</i>	Epidemic typhus Recrudescent typhus (Brill-Zinsser disease)	Louse	Human ^b	++	Africa, South America
<i>R. typhi</i>	Endemic/murine typhus	Flea	Rodents	-	worldwide
<i>Orientia tsutsugamushi</i>	Scrub typhus	Mite ^a	Rodents	++	Far East

^aVertically transmitted in arthropod.

^bNon-human vertebrates are possibly also involved.

II. Spotted fever^c

Organism	Disease	Arthropod vector	Vertebrate reservoir	Clinical severity	Geographic distribution
<i>R. rickettsii</i>	Rocky Mountain spotted fever	Tick ^a	Dogs, rodents	+	Rocky Mountain states, eastern USA
<i>R. akari</i>	Rickettsial pox	Mite ^a	Mice	-	Asia, Far East, Africa, USA
<i>R. conorii</i>	Mediterranean spotted fever	Tick	Dogs	+	Mediterranean
<i>Rickettsia japonica</i>	Oriental spotted fever	Tick/mite			
<i>Rickettsia australis</i>	Queensland tick typhus	Tick			Australia

^aVertically transmitted in arthropod

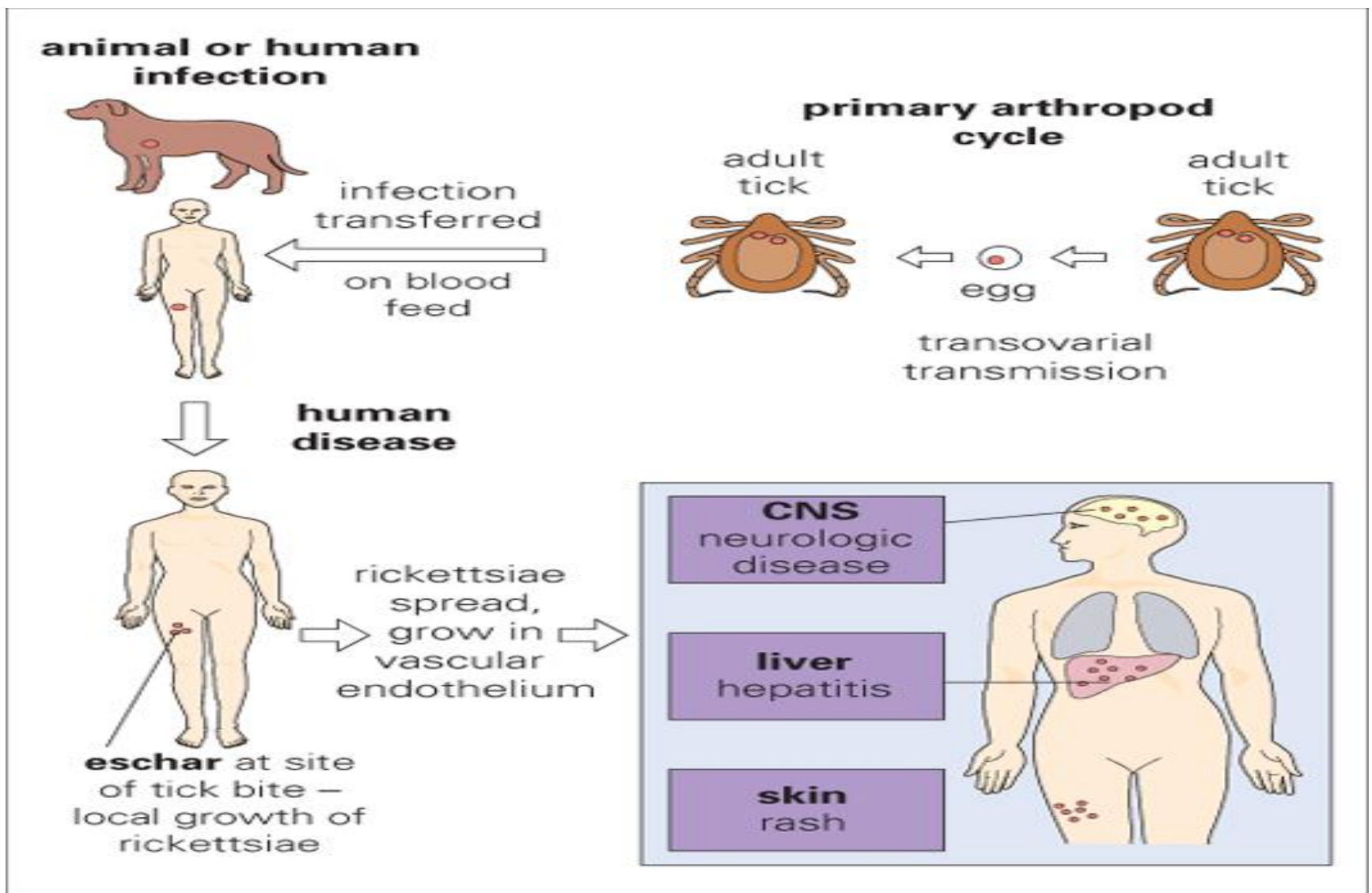
^cOther rickettsiae cause similar tick-borne fevers in Africa, India, Australia.

III. Others

Organism	Disease	Arthropod vector	Vertebrate reservoir	Clinical severity	Geographic distribution
<i>Coxiella burnetii</i>	Q fever/Atypical pneumonia	None	Sheep, goats, cattle	+	worldwide
<i>Bartonella henselae</i>	Cat-Scratch fever	none	Cats and dogs		
<i>Bartonella quintana</i>	Trench fever	Louse	Human	+	Asia, Africa, Central and South America ^d
<i>Ehrlichia chaffeensis</i> ^e	Fever (ehrlichiosis)	Tick	?	+	USA, Japan (<i>E. sennetsu</i>)

^dMultiply extracellularly; 1 million soldiers infected in the First World War.

^eIsolated at Fort Chaffee, Arkansas; parasitizes lymphocytes, monocytes, neutrophils.



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Typical events in rickettsia infection. There is no direct person-to-person spread. Q fever is atypical (see text). Typhus is unusual because the infected arthropod transmits from person to person, eventually dies and there is no eschar. CNS, central nervous system

Clinical features

Rickettsiosis

- Characteristic triads of symptoms include **fever, headache and rash**
- Rash spread from the trunk to the extremities (centrifugal) is typical for **typhus group** and MR 10-60%
- Rash spread from the extremities to the trunk (centripetal) is typical for **spotted fever** group

Q fever *Coxiella burnetii*

- No rash with Q fever
- Self limited febrile illness, atypical pneumonia and hepatitis
- Chronic form: Endocarditis, granuloma formation in liver, bone marrow and spleen

Laboratory diagnosis

- **Microscopic examination** of cutaneous biopsy materials from rash:
DFA stain
- **Culture:**
Cell/Tissue culture: 4-7 days
Embryonated hen's egg (yolk sac)
- **Serology: commonly used**
Weil-Felix test- (WF): antibody against antigens of *P. vulgaris* OX19, OX2 (Typhus group) or *P. mirabilis* OXK (scrub typhus group)
False positive and negative results common
- **Molecular methods (PCR):**
Amplification of Rickettsia DNA from clinical samples

Interpretation of Weil-Felix test

	Ab-OX 19	Ab-OX2	Ab-OXK
Epidemic typhus	++	<u>+</u>	-
Endemic typhus	++	-	-
RMSF	+	+	-
Scrub typhus	-	-	+
Q fever	-	-	-

Weil-Felix Reaction:

Patients with rickettsia infection have antibodies against Rickettsia. When patient serum is mixed with Proteus antigens, anti-rickettsial antibodies cross-react and agglutinate (Weil-Felix is negative in Coxiella infection).

Disease	Weil-Felix		
	OX-19	Ox-2	OX-K
Rocky Mountain spotted fever	+	+	-
Rickettsial pox	-	-	-
Epidemic typhus	+	-	-
Endemic typhus	+	-	-
Brill-Zinsser disease	+1-	-	-
Scrub typhus	-	-	+
Trench fever	-	-	-
	-	-	-

Treatment

- Tetracycline or chloramphenicol or doxycycline or TMP-SXT or rifampin

Prevention

- **Personal hygiene**
- Arthropod and rodent control are possible but difficult
- Pasteurization milk for Q-fever

3. PLAGUE

Yersinia pestis

Yersinia pestis

- Gram negative non-motile, non-spore-forming short rods
- Non-lactose fermenter (NLF), facultative anaerobes
- Bipolar staining appearance
- Causative agent of plague (Zoonotic disease-rodents)
- "Black death" 25 million (1/4 of population) people died in Europe

Virulence factors

- LPS: Endotoxin
- Yersinia Outer Membrane Proteins (YOPs)
Resist serum bactericidal activity
- Fibrolysin, coagulase helps the bacteria to survive in the vector
(rat flea, *Xenopsylla cheopis*)
- V-W ags
- Capsular protein, F1:

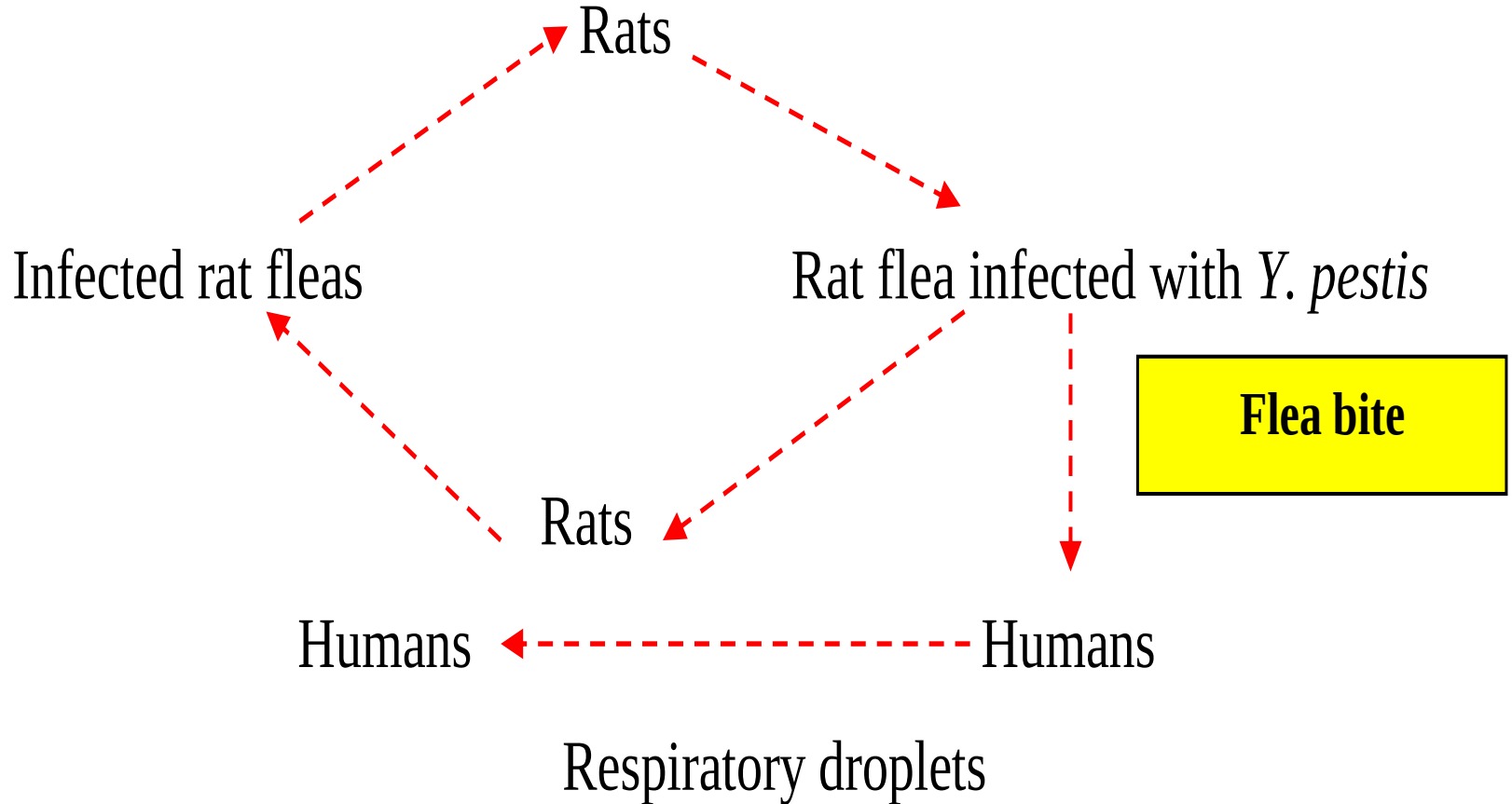


Antiphagocytic properties

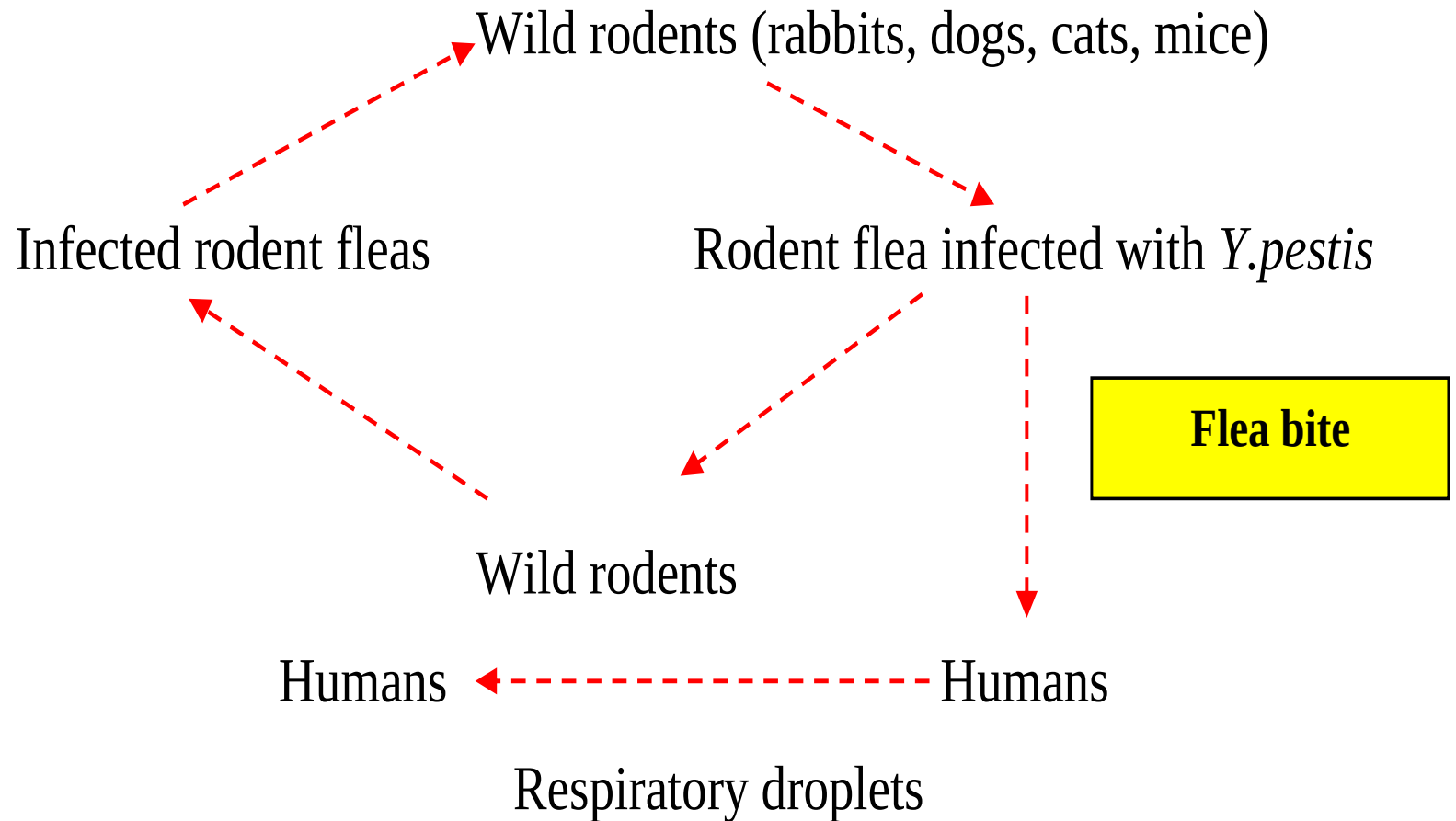
Multiply intracellularly -macrophages

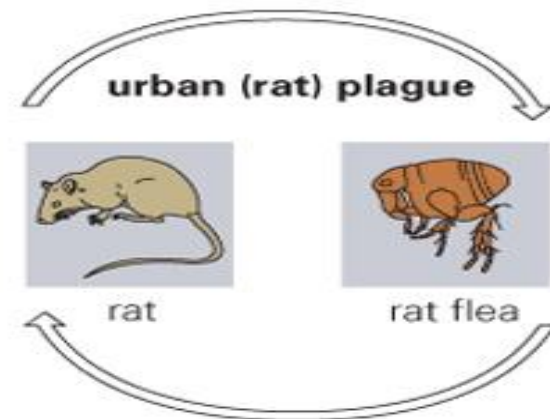
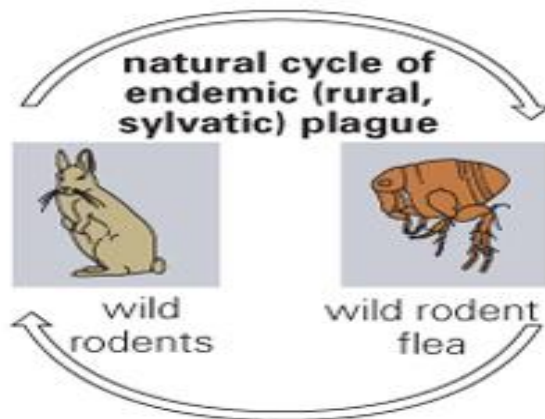
Epidemiology

- Urban (domestic) Plague: Epidemic type



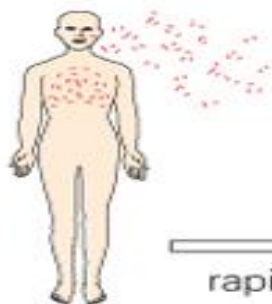
- Sylvatic (wild) plague: Sporadic type





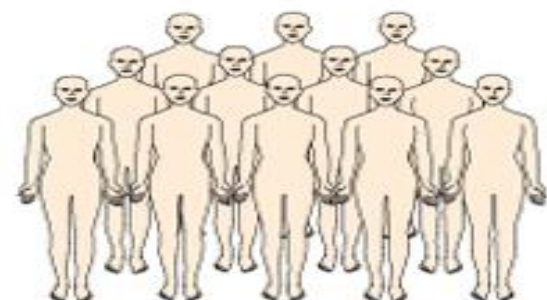
bubonic plague
no person-to-person spread

if there is profuse bacterial growth in lung

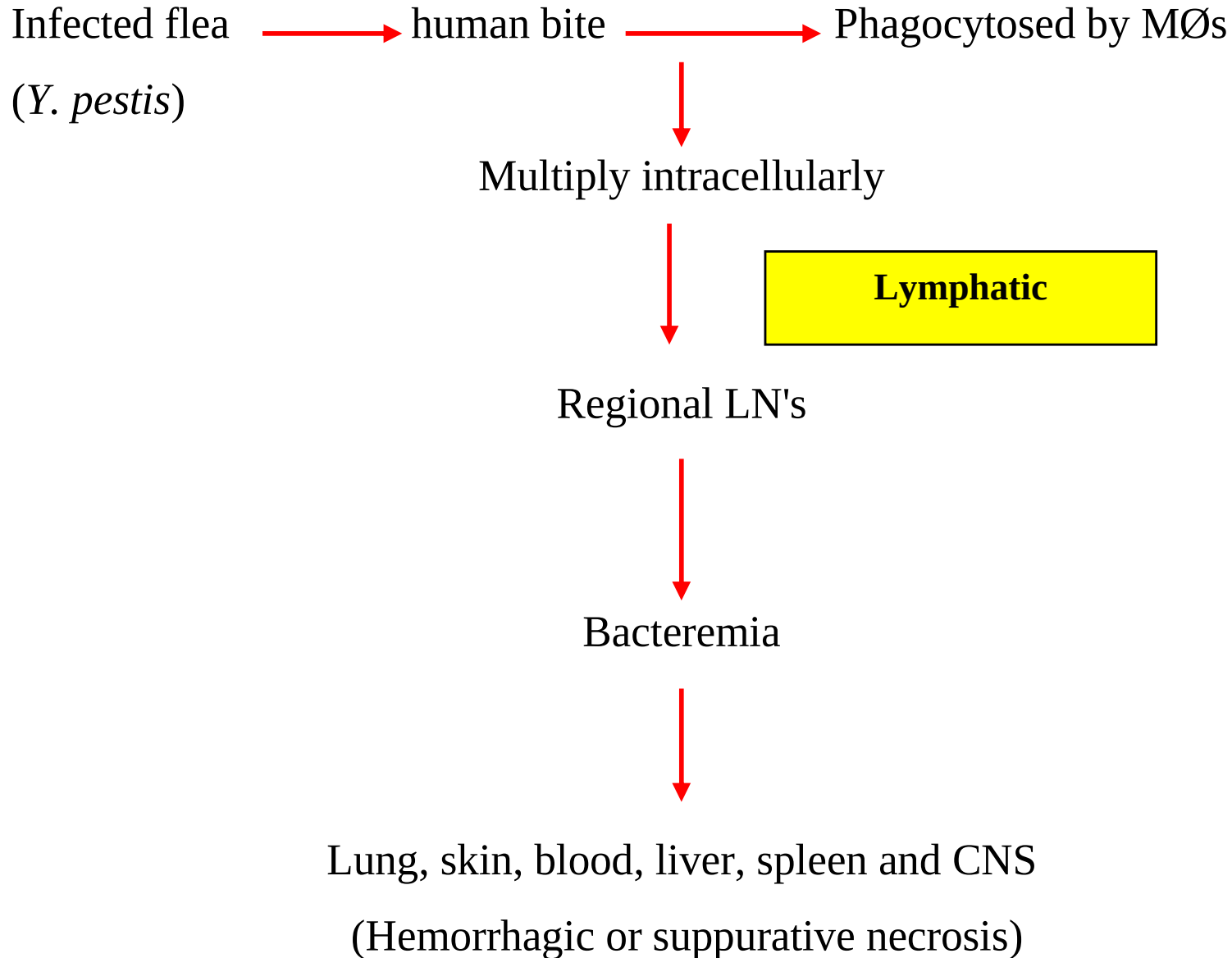


pneumonic plague

rapid person-to-person spread



Pathogenesis and pathology



Clinical features: Plague

Bubonic plague

- Fever, painful swelling of axillary and inguinal lymph nodes (bubo)
- MR: 70-75%

Pneumonic plague

- Fever, cough, hemoptysis with or without bubo
- Cyanosis-death
- MR: 100%

Septicemic plague

- Endotoxic shock

Cutaneous plague

- Pustule, carbuncle, ecthyma gangrenosum

Meningitis

Laboratory diagnosis of plague

Specimen collection and Transport

- Lymph node aspirate from bubo
- Sputum
- CSF
- Blood: septicemia

Microscopic examination

- Gram/Giemsa/Wright stain

Bipolar stained cocco-bacilli (safety pin appearance)



Dark stained bipolar ends of *Yersinia pestis* can clearly be seen in this Wright's stain of blood from a plague victim

Culture: Blood agar/enteric culture media

Serology Increase antibody titer- in convalescent sera-diagnostic

Treatment

- Streptomycin
- Tetracycline/doxycycline/chloramphenicol/TMP-SXT as alternatives

Prevention

- Chemoprophylaxis: Tetracycline close contacts
- Travel restriction
- Rodent control
- Killed vaccine: risk groups (travelers to hyperendemic areas)
Formalin-killed bacteria and gives partial protection
- Isolation of infected/suspected patients: Pneumonic plague

VECTOR BORNE VIRAL DISEASES

ARBOVIRAL DISEASES

Arboviruses

Arthropod-Borne Viruses (Arboviruses)

General Properties

- 450 arboviruses: 100 pathogenic to humans
- Transmitted to humans via blood sucking arthropods bite (mosquitoes, sand flies, ticks)

Examples of Arthropod Vectors



Aedes Aegypti



Assorted Ticks



Culex Mosquito



Phlebotomine Sandfly

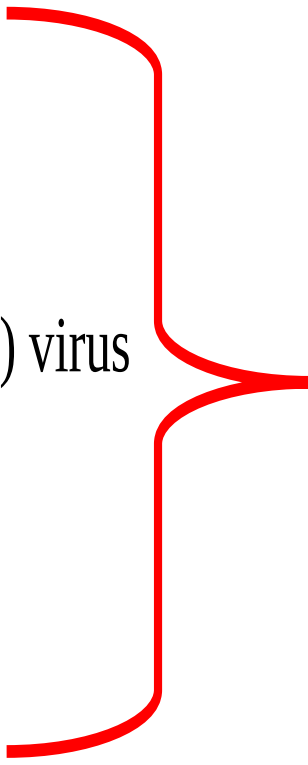
Virus family	Type	Envelope	Size (nm)	Nucleocapsid Symmetry
Togaviridae	SS-RNA +ve sense Linear	+	60-70	Cubic/Icosahedra
Flaviviridae	SS-RNA +ve sense Linear	+	40-60	Cubic/Icosahedra
Bunyaviridae	SS-RNA -ve sense	+	80-120	Helical/Cylindrical
Reoviridae Other than Rotavirus	DS-RNA linear (segmented)	-	60-80	Cubic/Icosahedra

Classifications

I. Togaviridae

Alphaviruses

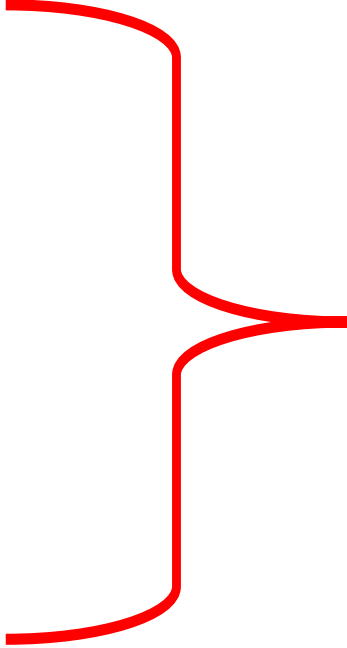
- Western Equine Encephalitis (WEE) virus
- Eastern Equine Encephalitis (EEE) virus
- Venezuelan Equine Encephalitis Viruses (VEE) virus
- Chikungunya virus
- Sindbis virus
- Semliki Forest virus



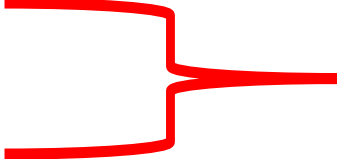
Alphaviruses
Vector: Mosquito

I. Flaviviridae

- Yellow fever virus
 - Dengue fever virus (4 serotypes)
 - West Nile fever virus
 - St. Louis encephalitis virus
 - Japanese B encephalitis virus
 - Zika virus
-
- Russian spring summer encephalitis virus
 - Powassan virus



Flaviviruses
Vector: Mosquito



Flaviviruses
Vector: Ticks

I. Bunyaviridae

- California encephalitis virus



- Sand fly fever virus



- Rift valley fever virus



- Crimean-Congo-Hemorrhagic fever virus



I. Reoviridae

- Colorado tick fever virus



Natural host (reservoirs): arboviruses

- **Birds** EEE, WEE & St. Louis encephalitis & Japanese encephalitis V, West Nile encephalitis viruses
- **Pigs** Japanese encephalitis virus
- **Monkeys:** Jungle Yellow Fever & Dengue Fever virus
- **Rodents:** VEE, Russian Spring-Summer encephalitis California encephalitis viruses
- **Vectors/Ticks:** Trans-ovarian transmission of virus common between arthropod vectors

Vectors

- **Mosquitoes**

- Japanese encephalitis V, Dengue fever V, Yellow fever V, St. Louis encephalitis V, EEE-V, WEE-V, VEE-V, Rift Valley fever virus, West Nile encephalitis V, California encephalitis virus etc.

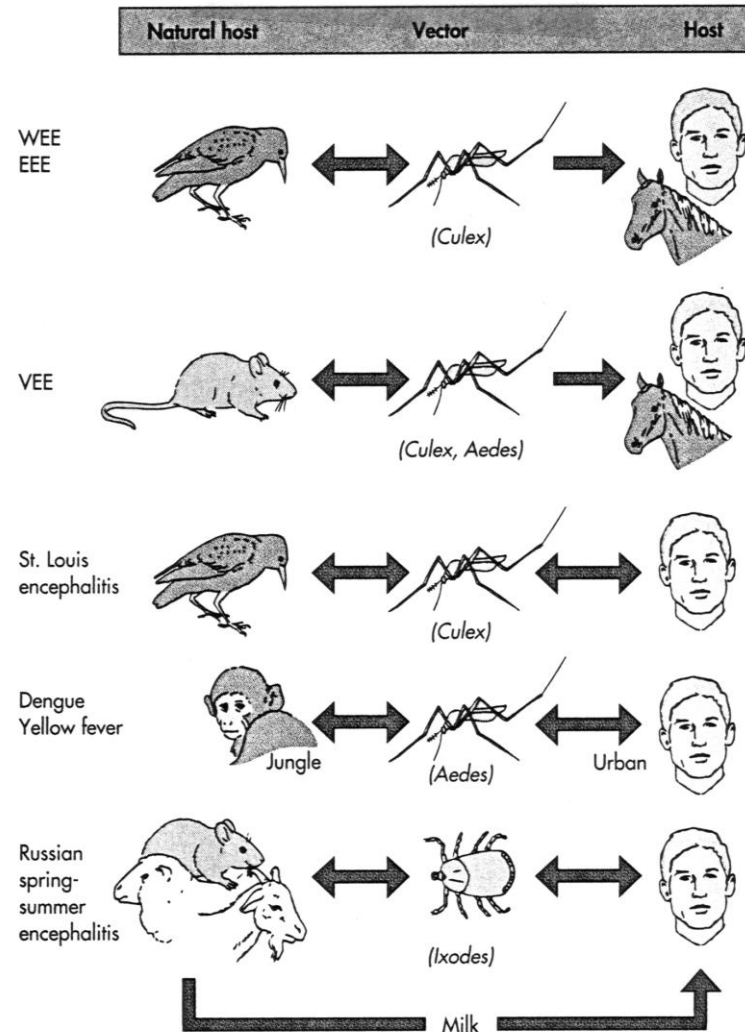
- **Ticks**

- Crimean-Congo hemorrhagic fever V, Colorado tick fever V, Powassan V etc.

- **Sand fly**

- Sandfly fever virus

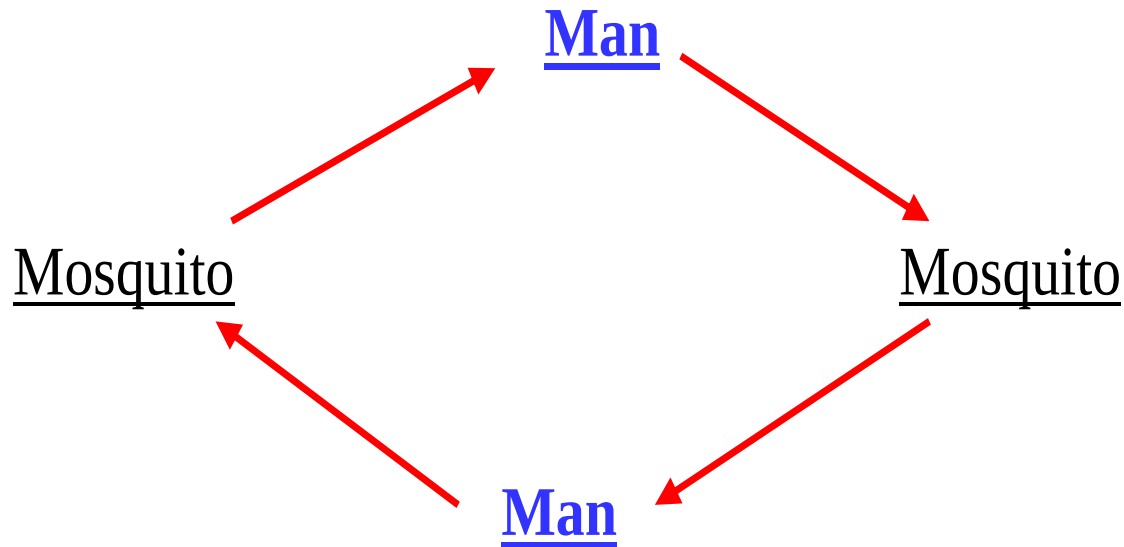
Fig. 63-5 Patterns of Alphavirus and Flavivirus Transmission



From Murray et. al., Medical Microbiology
5th edition, 2005, Chapter 63, published by
Mosby Philadelphia.,

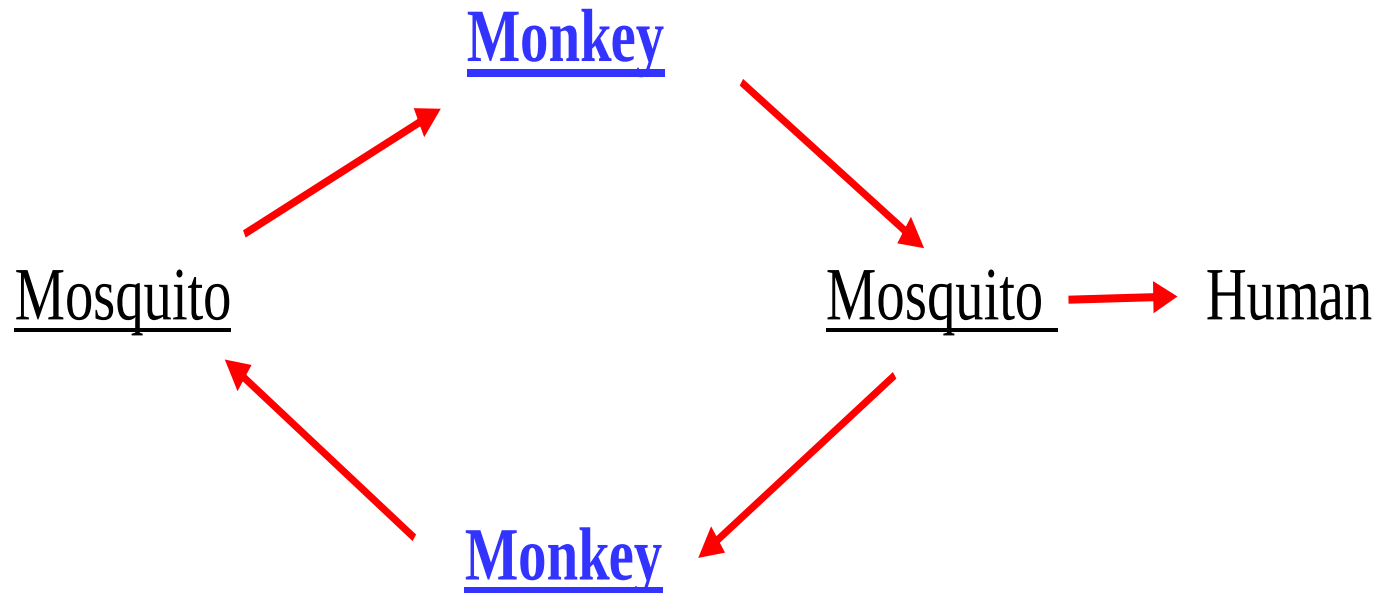
Epidemiology of Arboviral Diseases

I. Urban cycle



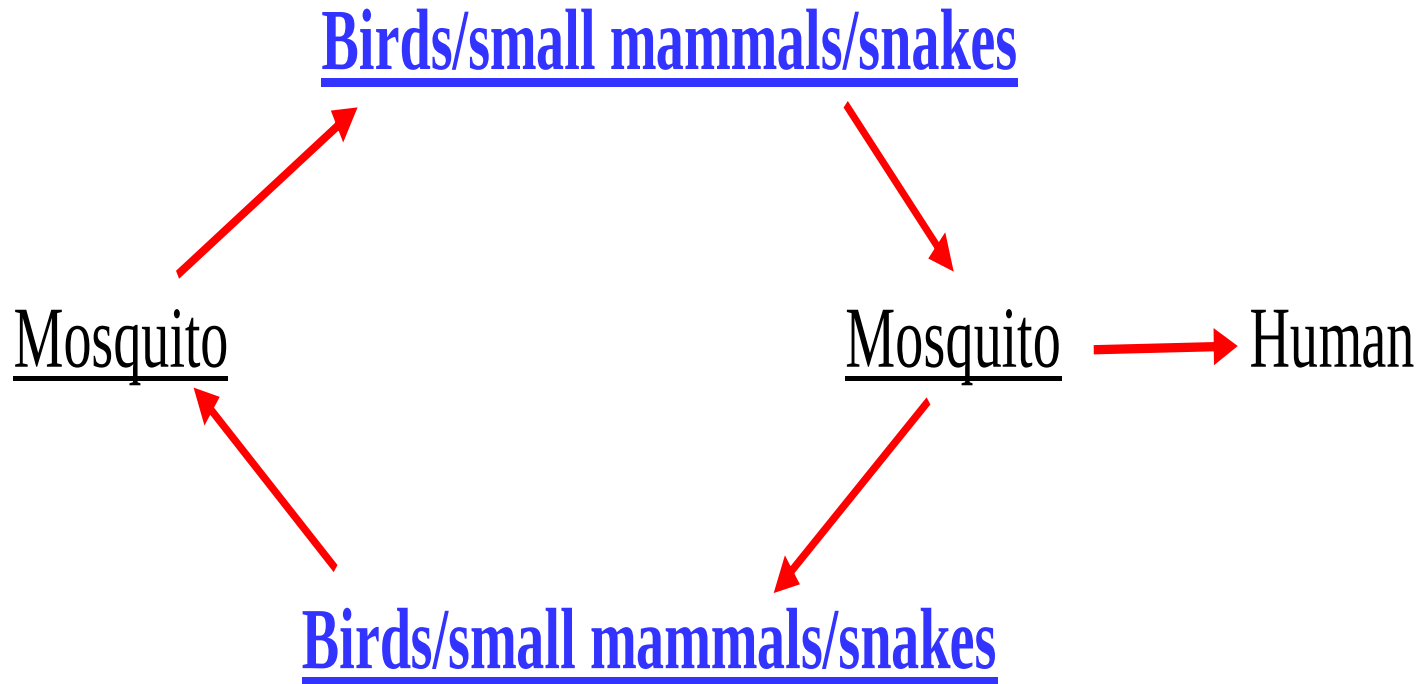
e.g. Urban Yellow fever
Urban Dengue fever
St. Louis encephalitis

II. Sylvatic (Jungle Cycle)



e.g. Jungle Yellow fever

Jungle Dengue fever



e.g. Western equine encephalitis (WEE)

Eastern equine encephalitis (EEE)

Pathogenesis and Pathology of Arboviral Diseases

Arboviruses- arthropod bite



Viremia

, spleen, bone marrow, lymph nodes + kidney + CNS + vascular endothelium



Inflammation, hemorrhage, necrosis and degeneration

Clinical Features of Arboviral Infections

1. Encephalitis/aseptic meningitis/meningoencephalitis

Case-fatality rates average 10 – 20 % (higher in elderly)

2. Hemorrhagic fever

Yellow fever (YF): Urban and Jungle

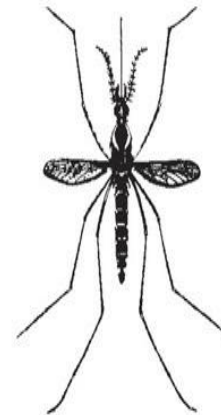
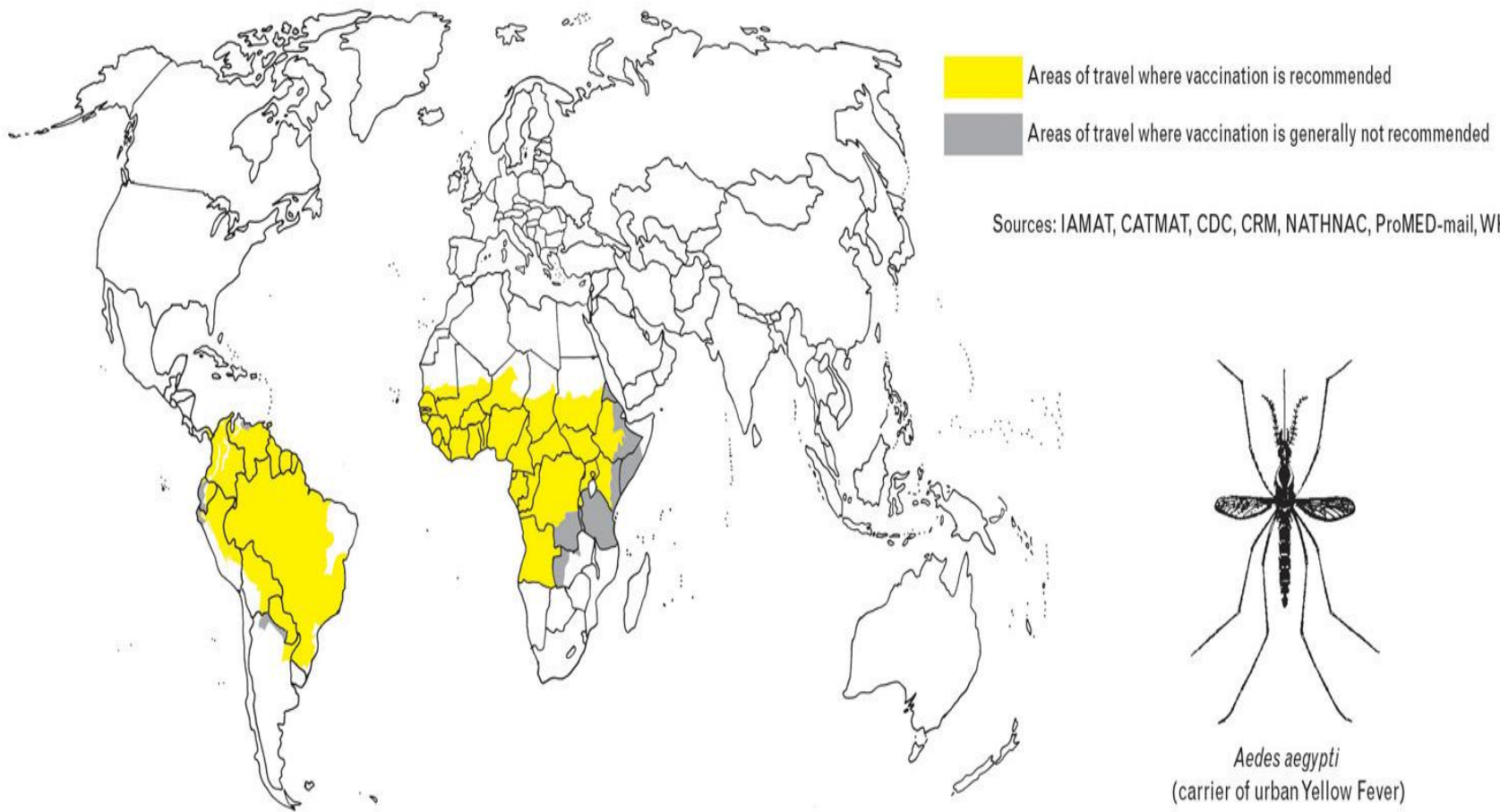
- GD: Tropical belt countries-endemic

Central and South America

Caribbean countries

Africa: Ethiopia, Sudan, Nigeria, etc....

- IP: 3-6 days
- Target organs: LN's, **liver**, spleen, heart, gut, and kidney



Aedes aegypti
(carrier of urban Yellow Fever)

Clinical features of YF

- Chills, fever, headache, nausea, vomiting, jaundice, hemorrhagic manifestations (petechial skin lesions, epistaxis, GI hemorrhage-black vomit)

Recovery 50%: without chronicity-liver

Early symptoms of yellow fever include



Sudden onset of fever



Chills



Severe headache



Back pain



General body aches



Fatigue and Weakness



Nausea and Vomiting

Severe cases include



High Fever



Bleeding and Shock



Yellow discoloration of the skin and the whites of the eyes

Dengue fever (Break Bone Fever)

- Urban and Jungle dengue fever
- GD: Common in South East Asia, the Caribbean and Central and South America, Africa
- IP: 2-7 days
- Target organs: RES + other organs may be involved such as the heart, lungs and GI tract
- Fever, hemorrhagic manifestations, hepatomegaly, thrombocytopenia-shock death
- MR: 5-10%

Symptoms of Dengue fever

Febrile phase

sudden-onset fever

headache

mouth and nose
bleeding

muscle and
joint pains

vomiting

rash

diarrhea

Critical phase

hypotension

pleural effusion

ascites

gastrointestinal
bleeding

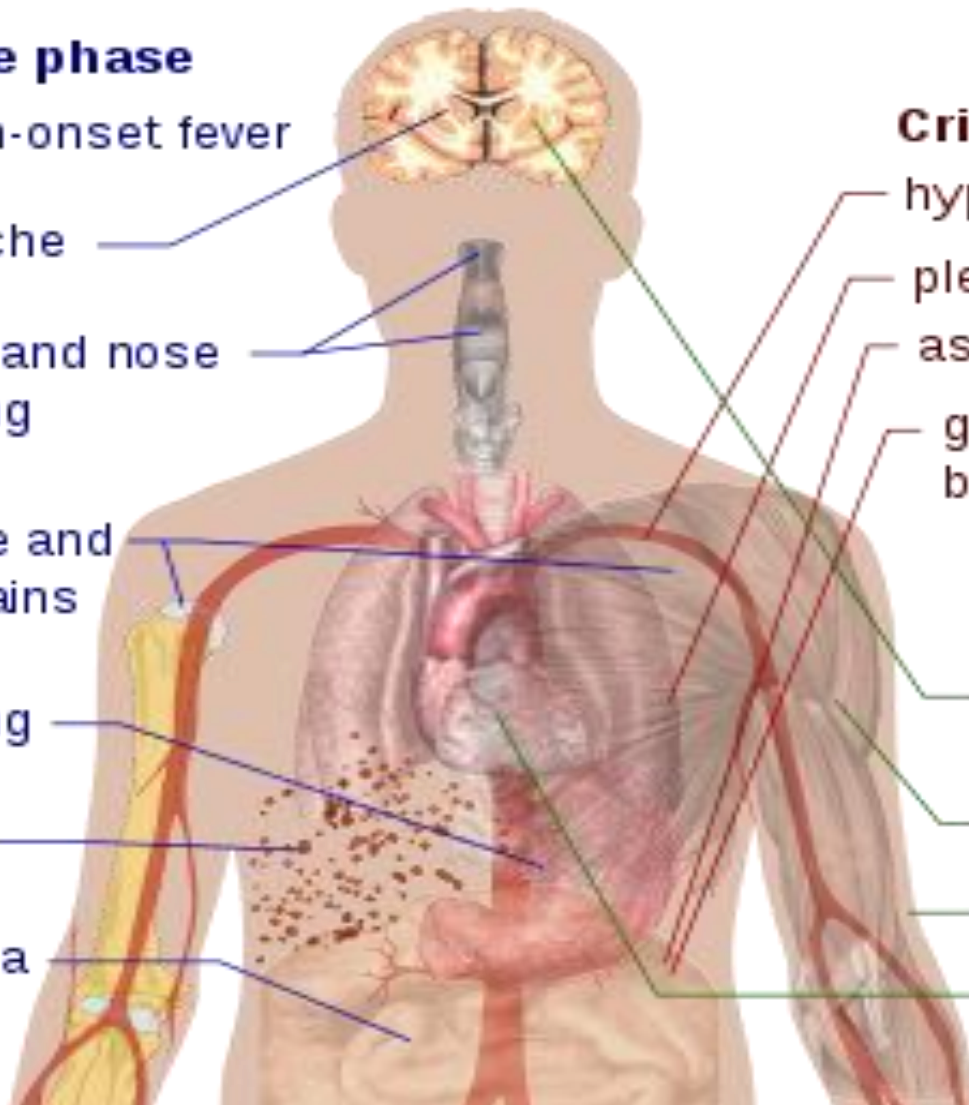
Recovery phase

altered level of
consciousness

seizures

itching

slow heart rate



Other Hemorrhagic Fever caused by arboviruses

- Rift valley Fever
- Sand fly fever
- Crimean-Congo hemorrhagic fever
- Colorado Tick fever
- Ross Fever

Summary of Arboviral Diseases

- Acute febrile illness
- Hemorrhagic fever
- Aseptic meningitis
- Encephalitis/meningoencephalitis
- Hepatitis

Diagnosis of arboviral diseases

- Clinical
- History of recent travel to endemic areas/contact rodents
- Culture
- Serology

Treatment and Prevention

- Supportive
- Restriction of travel
- Vector/rodent control
- Vaccine before travel to endemic areas

Yellow fever (live attenuated vaccine 17D strain)

Effective and safe

Single dose protective for 10 years

Booster dose every 10 year

Arboviruses causing encephalitis/ aseptic meningitis

Arboviruses	Geographic distribution	Vector for human infection	Vertebrate reservoir	Severity of infection
Easter equine encephalitis V (Togavirus)	USA (Atlantic Gulf states)	Aedes spp. mosquitoes	Wild birds, horses (dead-end hosts)	50% case fatality
Western equine encephalitis V (Togavirus)	USA (west of Mississippi)	Culex spp. mosquitoes	Wild birds, horses (dead-end hosts)	Up to 2% case fatality
West Nile encephalitis V (Flavivirus)	Africa, Europe, Central Asia, USA	Culex spp. mosquitoes	Birds	Up to 5% case fatality
St Louis encephalitis V (Flavivirus)	USA (Southern, Central and Western States)	Culex spp. mosquitoes	Wild birds	10% case fatality
California encephalitis V (Bunyavirus)	USA (Northern and Central States)	Aedes spp. mosquitoes	Small mammals	Fatalities rare
Japanese encephalitis V (Flavivirus)	Far East, South East Asia	Culex spp. mosquitoes	Birds, pigs	8% case fatality
Murray valley encephalitis V	Australia	Culex spp. mosquitoes	Birds	Up to 70% case fatality
Tick-borne encephalitis V	Eastern Europe	Tick	Mammals, birds	Up to 10% case fatality (variable)
Venezuelan encephalitis V (Togavirus)	Southern USA, Central and South America	Mosquito	Rodents	70% case fatality (cases rare)
Powassan V (Flavivirus)	USA, Canada	Tick	Rodents	Cases rare

Viruses	Disease	Geographic distribution	Vector	Animal reservoir
Yellow fever (flavivirus)	Fever, hepatitis	Africa, Central and South America	Mosquito <i>Aedes</i> spp.	Nil (monkeys for jungle type)
Dengue (4 serotypes) (flavivirus)	Fever, rash (hemorrhagic shock syndrome)	India, southeast Asia, Pacific, South America, Caribbean	Mosquito	Nil
Kyasanur forest (flavivirus)	Hemorrhagic fever	India	Tick	Monkeys, rodent
Ross river (alphavirus)	Fever, arthralgia, arthritis	Australia, Pacific Islands	Mosquito	Birds
Rift Valley fever (bunyavirus)	Fever, sometimes hemorrhage	Africa	Mosquito	Sheep, cattle, camels
Sandfly fever (bunyavirus)	Fever (mild disease)	Asia, South America, Mediterranean	Sand flies	Gerbils
Congo-Crimean hemorrhagic fever (bunyavirus)	Fever, hemorrhage	Asia, Africa	Tick	Rodents
Colorado tick fever (reovirus)	Fever, myalgia	USA (Rocky Mountains)	Tick	Rodents
La Crosse (bunyavirus)	Fever	USA	Mosquito	Rodents